

What's New in In-Circuit Testing

In-circuit testers are transforming into all-in-one electrical test controllers. The latest ICTs can be used for not only manufacturing defects but also to test functional failures, and even parametric testing

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In-circuit testers (ICTs) have become more versatile and efficient as these now come loaded with additional electrical test capabilities and functionalities. The most capable ones have in-circuit test, boundary scan, functional test as well as programming techniques integrated into a single platform.

What drove these changes

The inability of any single test methodology to completely test circuit assemblies is one reason. In addition, the constant drive from circuit assemblers to reduce test costs is driving traditional ICT systems to take the shape of more holistic test systems.

Newer ICT systems can be better described as generic all-in-one electrical test controllers rather than just ICT systems. This is because many of these

have different forms of ICT including automated probing, optical inspection and boundary scan combined with functional testing of circuit assemblies.

Let us take a look at the latest changes in ICT systems, and what they deliver to the users.

Small-footprint automated systems moving to 'zero' footprint

With the introduction of zero-footprint designs incorporating multi-core test capability combined with automated material handling, the latest inline systems have taken a step forward in reduction in the cost of tests. These systems are a boon to manufacturers that want to increase their capability but do not have the additional real estate to do so.

The industry's previous solution to this has been use of many discrete automation methods including robotic helps, automatic probes and semi-

automatic press for PCB testing along with large automated ICT systems. However, all these systems contribute to an inefficient use of factory space and low machine utilisation due to size of conveyors and speed of travel. Moreover, the additional cost of moving the product from one equipment to another is always a concern.

The newer inline ICT systems introduced last year come with small footprint and there is no manual handling involved. You can integrate your ICT into the assembly line and there is no need of any operator handling.

The small footprints in inline systems are realised by using compact test heads relocated closer to the device under test (DUT). This enables ICT to fit easily within the narrowest automated lines, without the need for any additive storage bay or incremental factory space, thus saving additional cost as well as labour.

Teradyne's TestStation Multi-Site

Inline ICT system uses a novel 'zero-footprint' test head design, which fits entirely within the narrowest automations. Teradyne also has offline counterparts with sub 1 m2 footprint and 2560 test-points capability which help PCB manufacturers free up factory space.

Agilent's i1000D ICT system is another small-footprint design that unifies offline and inline systems onto the same design. Agilent offers easy upgrades to convert offline to inline systems so you can re-use the same system.

Spea's 3030 system is another true multi-core tester offering up to 4x

Latest ICT Solutions

Module	Manufacturer	Key features
Teststation Multi-Site	Teradyne	'Zero-footprint' design using up to 70% less factory space and power provisioning - Delivers full parallel test of multiple boards - Equivalent productivity of two or more conventional MDA/ICT systems
QTtouch 1248	QmaxTest	- Able to probe 2 test points as close as 20 mils/500 microns with unique angular probing - Learn and compare VI curve including Automatic 'best fit curve' - Electrical measurements of components
TR5001T SII TINY	Test & Research India Pvt Ltd	- Compact and lightweight full-featured ICT solution - Parallel testing using multiple USB-connected units - Programmable DUT power supplies
Focus 2000	Kyoritsu Electric India Pvt Ltd	- Fastest manufacturing defect analyser (MDA) - Ultra-advanced solid-state switching technology - Easy maintenance
Flying Scorpion FLS9800xi	Acculogic	- Full-function double-sided system - Programmable probe angle +6° to -6° degrees - Closed-loop AccuFast drive system with ±0.1 micron resolution
Flying Probe S2	Spea	- Manual + automated + in-line board loading combined into the same system - Highest measurement performance and accuracy (0.1pF) 4x parallel fast on-board programming
Medalist i1000D	Agilent	'Small-footprint' design unifying offline and inline systems - Low-cost fixturing requirements and superior signal quality - Control path for board alignment verification ensuring board safety